

TABLE I
SOME SDRs AVAILABLE ONLINE

SDR	Tuning frequency (minimum)	Tuning frequency (maximum)
Airspy	24MHz	1.8GHz
BladeRF	300MHz	3.0GHz
FUNcube Pro+	150kHz 410MHz	260MHz 2.05GHz
HackRF	30MHz	6GHz
RTL-SDR (R820T)	24MHz	1.7GHz
RTL-SDR V3	500kHz	1.7GHz
SDRPlay	100kHz	2GHz

An external antenna is connected to the SDR. Length of the antenna depends on the distance between the computer and transmitting station.

Another aspect of DRM reception is suitable software. Some free and downloadable software to receive



Fig. 4: Software-defined radios

TABLE II
DIGITAL TRANSMISSION OF AIR IN MEDIUM WAVE

(A) OPERATION IN PURE DIGITAL MODE

Location/station	Duration	Frequency	Programme content	
			First channel	Second channel
Chennai	05:55-17:05 18:00-23:10	783	Vividh Bharati (Chennai)	FM Rainbow (Chennai)
Delhi	05:55-11:00 12:00-17:30 18:15-23:10	1368	Vividh Bharati (Delhi)	FM Rainbow (Delhi)
Rajkot	12:00-13:45	1071	Urdu	

(B) OPERATION IN SIMULCAST MODE

(Transmitters scheduled to carry out both analogue and digital transmissions as per timings of existing analogue operations)

Location/station	Frequency (kHz) in Pure Digital	Content of Digital Program
*Ahmedabad	855	Rainbow India
*Ajmer	612	Rainbow India
*Bengaluru	621	FM Gold
*Barmer	1467	VBS Mumbai
*Bikaner	1404	VBS Mumbai
*Chennai-A	729	Ragam Channel
Chinsurah	603	Rainbow India
*Delhi-A	828	FM Gold
*Dharwad	774	Rainbow India
Dibrugarh	576	VBS Mumbai
*Guwahati-B	1044	Rainbow India
*Itanagar	684	Rainbow India
*Jabalpur	810	Rainbow India
Jalandhar	882	FM Gold
**Jammu	999	Rainbow India
*Kolkata-A	666	FM Gold (Kolkata)
*Kolkata-B	1017	FM Rainbow (Kolkata)
*Lucknow	756	FM Gold
*Mumbai-A	1053	DTH Hindi
*Mumbai-B	567	DTH Marathi
*Panaji	1296	FM Gold
Passighat	1071	VBS Mumbai
*Patna	630	Rainbow India
*Pune	801	Rainbow India
*Rajkot	819	Rainbow India
Rajkot	1080	VBS/URDU
Ranchi	558	Rainbow India
*Siliguri	720	Rainbow India
Suratgarh	927	VBS Mumbai
Tawang	1530	VBS Mumbai
*Tiruchirappalli	945	VBS Chennai
*Varanasi	1251	VBS Mumbai
*Vijayawada	846	FM Gold

Recently, Hyderabad, Andhra Pradesh was added to the list.

* Operates on respective analogue frequency (digital frequency is 9kHz) in pure DRM mode between 15:30 and 16:30 (IST), every day except Sunday, carrying two channels.

** Operates on 990kHz (999kHz-9kHz) in pure DRM mode between 10:30 and 11:30 (IST), every day except Sunday, carrying two channels.